Whitepaper

UWB: The Decision-Maker's Guide

by Lubomír Mráz, UWB Evangelist and Co-founder of HID's Sewio

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Why Ultra-Wideband, Why Now

If you manage operations, efficiency is your currency. But you can't optimize what you can't see. In complex, fast-moving environments, visibility is often the first thing lost. That's where UWB comes in.

What is Ultra-Wideband Technology?

UWB gives devices precise **spatial awareness**. It enables detecting an object's location with centimeter-level accuracy, even in environments full of steel, concrete, and chaos. UWB is now widely known for smartphones and cars, but its real breakthrough came in factories and warehouses where top manufacturers use it to cut lead times, boost safety, and eliminate wasteful guesswork from their daily operations. The global UWB market grew from 109 million devices in 2019 to 3.6 billion in 2025 and continues to expand rapidly.

Ultra-Wideband, once a niche interest for tech enthusiasts, is now at a key turning point. The commercial sector is about to adopt UWB on a large scale, evaluating and integrating this technology. It's set to become a standard across the industrial sector in the near future.

Milan Šimek, Co-founder of HID's Sewio

Where is UWB used?

Shifting from Outdoors to Indoors

GPS changed the world when its precision signals became available to the public in 2000. Today, over half of all mobile apps rely on GPS: delivery services, fitness tracking, social media, and booking applications.

But GPS has one major flaw: it doesn't work indoors.

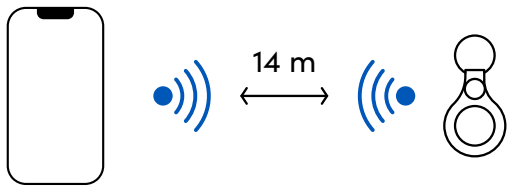
Just as GPS transformed outdoor navigation, UWB is transforming indoor positioning.

Concrete, steel, and industrial infrastructure block GPS signals, leaving factories and warehouses in the dark. That's a job for UWB.

It measures the distance between two devices (called peer-to-peer ranging), for example when you use your phone to find a key fob.

It can also pinpoint the exact location of an object with up to 10 cm accuracy when used with a real-time location system (infrastructure-based positioning).

Range between two devices



Absolute location

UWB Applications

UWB technology delivers significant value in both consumer and industrial markets. While both benefit from high accuracy and resistance to interference, the goals and technical implementations are different.

Consumer applications

In consumer applications, UWB is integrated into smartphones and wearables to enable precise, short-range interactions. Typical uses include:

- Digital car keys
- Smart home
- Location tags

These applications rely on peer-to-peer ranging, where two mobile devices communicate directly to measure distance.

Industry applications

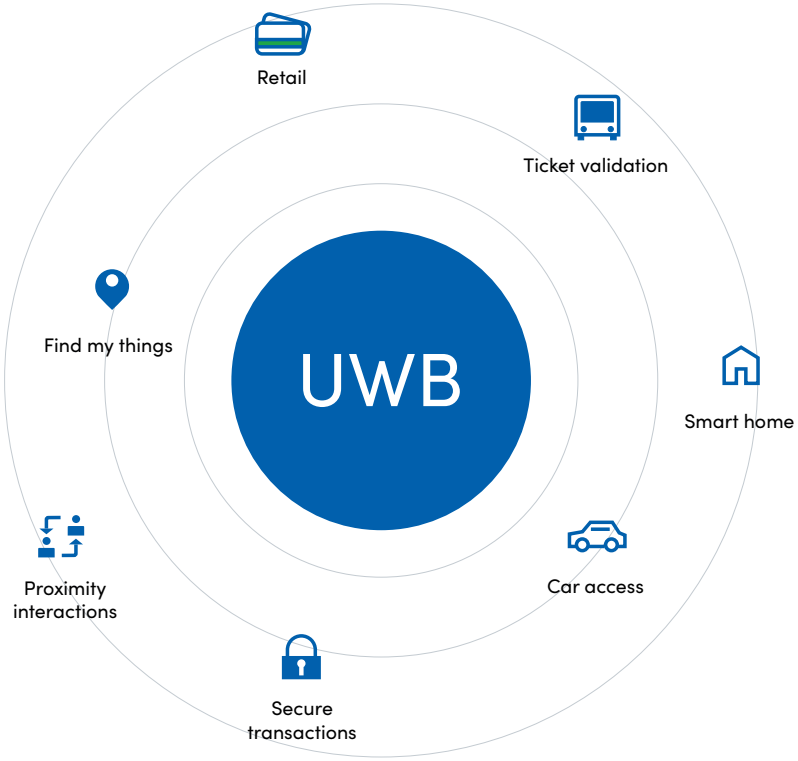
Industrial UWB applications rely on real-time location systems (RTLS) deployed across entire facilities. These systems use fixed infrastructure and mobile tags to provide continuous visibility of:

- Assets
- Material handling units
- Workers

UWB-powered RTLS enables better workflow management, improves safety, and supports automation in complex, harsh environments.

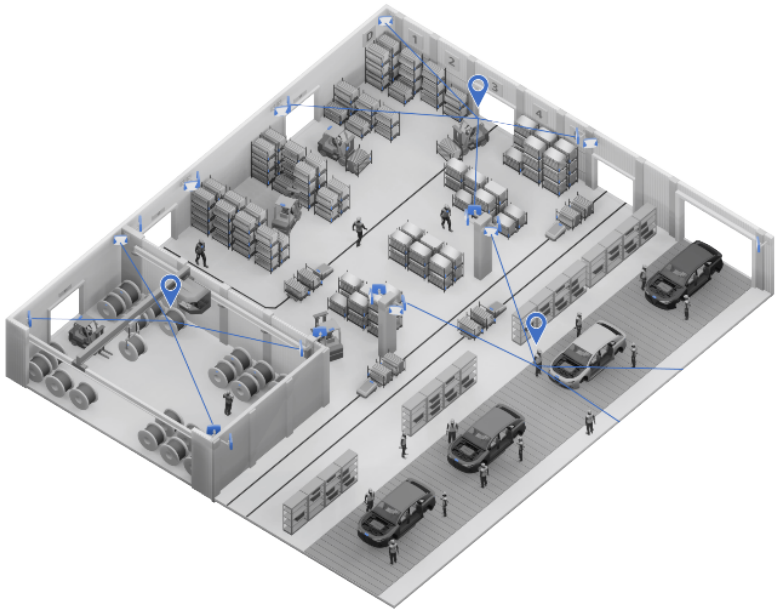
Phone-centric consumer applications

Peer-to-peer ranging



Industrial RTLS applications

Infrastructure-based location service



Tracking assets, material handling units, and workers provides real-time visibility, eliminates manual tasks, and improves workflow management, leading to greater efficiency and safety.

What Is a Real-Time Locating System (RTLS) and Why Use UWB?

RTLS provides continuous, real-time visibility of people, assets, and materials within a defined space. In industrial environments, this visibility unlocks automation, reduces manual tracking, and improves safety and efficiency.

To function, RTLS requires two main components:

Tags:

small devices attached to assets or worn by workers

Anchors:

fixed receivers across the space that read tag signals and communicate with the software

Among several technologies available for RTLS—such as BLE, RFID, Wi-Fi, and infrared—UWB stands out as the most accurate and reliable in industrial conditions.

Why UWB?

01 **Centimeter-level accuracy:** UWB offers location accuracy within 10–30 cm, significantly outperforming BLE and Wi-Fi in precision.

02 **Minimal signal interference:** Industrial spaces are full of metal, concrete, and moving machinery. UWB's wide bandwidth and short pulses let it cut through multipath interference where other signals fail

03 **High reliability at scale:** Whether it's tracking 50 forklifts or 2,000 pallets, UWB keeps performance steady, even in complex environments.

UWB isn't just a good fit for industry. It was built for it.

UWB's Unique Advantages

1. Precise positioning with high reliability

UWB's biggest strength. It measures the distances between objects with consistent, centimeter-level accuracy (up to 30 cm in industrial environments). This level of precision is made possible by UWB's wide bandwidth.

2. Industrial-grade accuracy that holds up in the real world

Some technologies can approach UWB's accuracy in ideal conditions, like offices or labs, but UWB consistently delivers over 99% accuracy even in tough industrial environments with metal, machinery, and interference. In settings where precise location is critical, UWB's proven "pallet-size" accuracy makes it the most reliable choice.

UWB solves issues such as:

- Misplaced assets and inventory errors
- Production delays from misrouted materials
- Compliance risks due to incorrect product traceability

Why does high accuracy matter for industrial shop floors?

Many technologies can locate objects within a meter, but their reliability is far lower than UWB's.

Other Radio Technologies

30% Position reliability

UWB

99% Position reliability

Aligning digital data with physical reality

The locations stored in your ERP system have to correspond to the actual locations of items in the production hall, especially if you need to ship orders quickly.

Technologies with low reliability can show the wrong location, cause delays, and extra work. That's why low-reliability positioning systems aren't suitable for industrial operations.

ERP system record

±3 m

Location technology
Narrowband

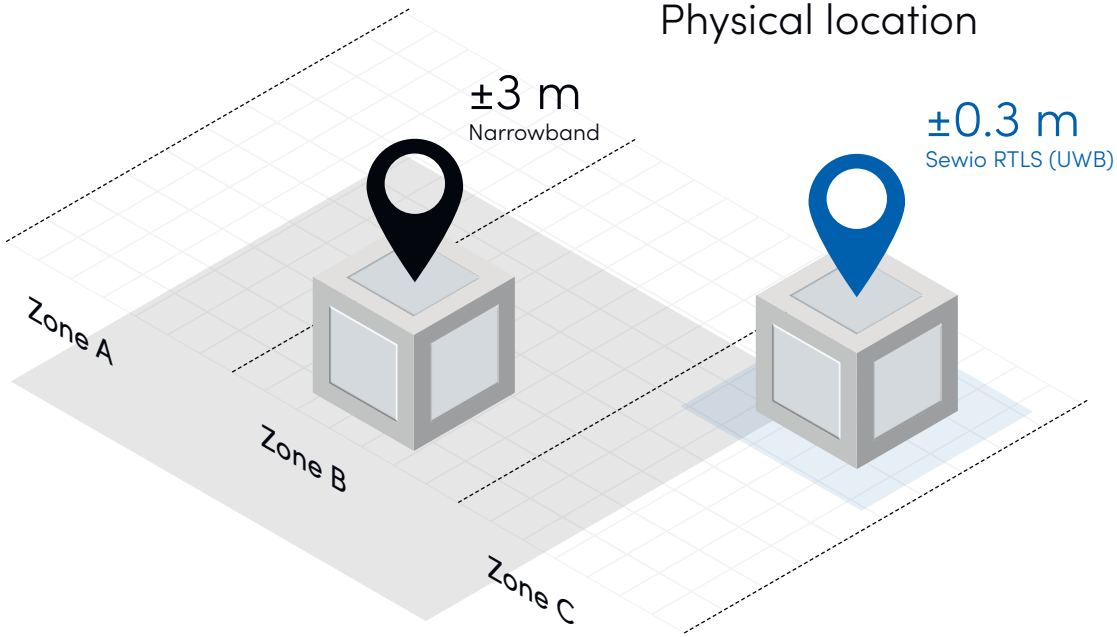
time	zone
10:32:22	A ✗
10:32:25	B ✓
10:32:28	A ✗
10:32:33	C ✗
10:32:37	B ✓

±0.3 m

Location technology
Sewio RTLS (UWB)

time	zone
10:32:22	C ✓
10:32:25	C ✓
10:32:28	C ✓
10:32:33	C ✓
10:32:37	C ✓

Physical location



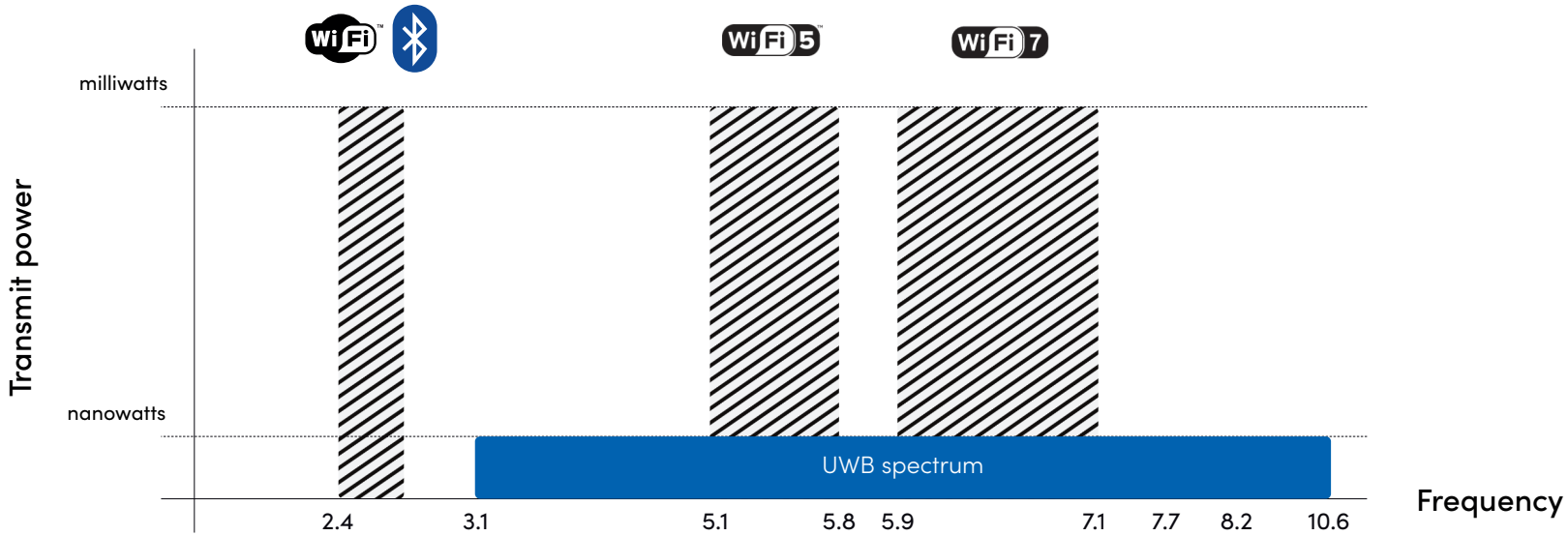
3. Built for coexistence and security

UWB's exceptionally low transmission power, roughly 100,000 times lower than that of other radio technologies, is one of UWB's greatest advantages.

It enables UWB to **operate without interference** in dense areas and next to other wireless technologies like Wi-Fi and Bluetooth. It also makes UWB ideal for **battery-powered devices**, as it needs less energy while maintaining high signal quality and precision.

Low power consumption offers two key advantages:

1. **Resistance to interference:** UWB performs next to other wireless technologies.
2. **Inherent security:** UWB transmissions are hard to detect, spoof, or intercept without specialized equipment.



How Does UWB Positioning Work?

The more bandwidth, the better the resolution.

Wide bandwidth is one of UWB's core strengths. The technology operates on unlicensed spectrum with bandwidths up to **500 MHz or even 1 GHz**—more than ten times that of technologies like Bluetooth or Wi-Fi.

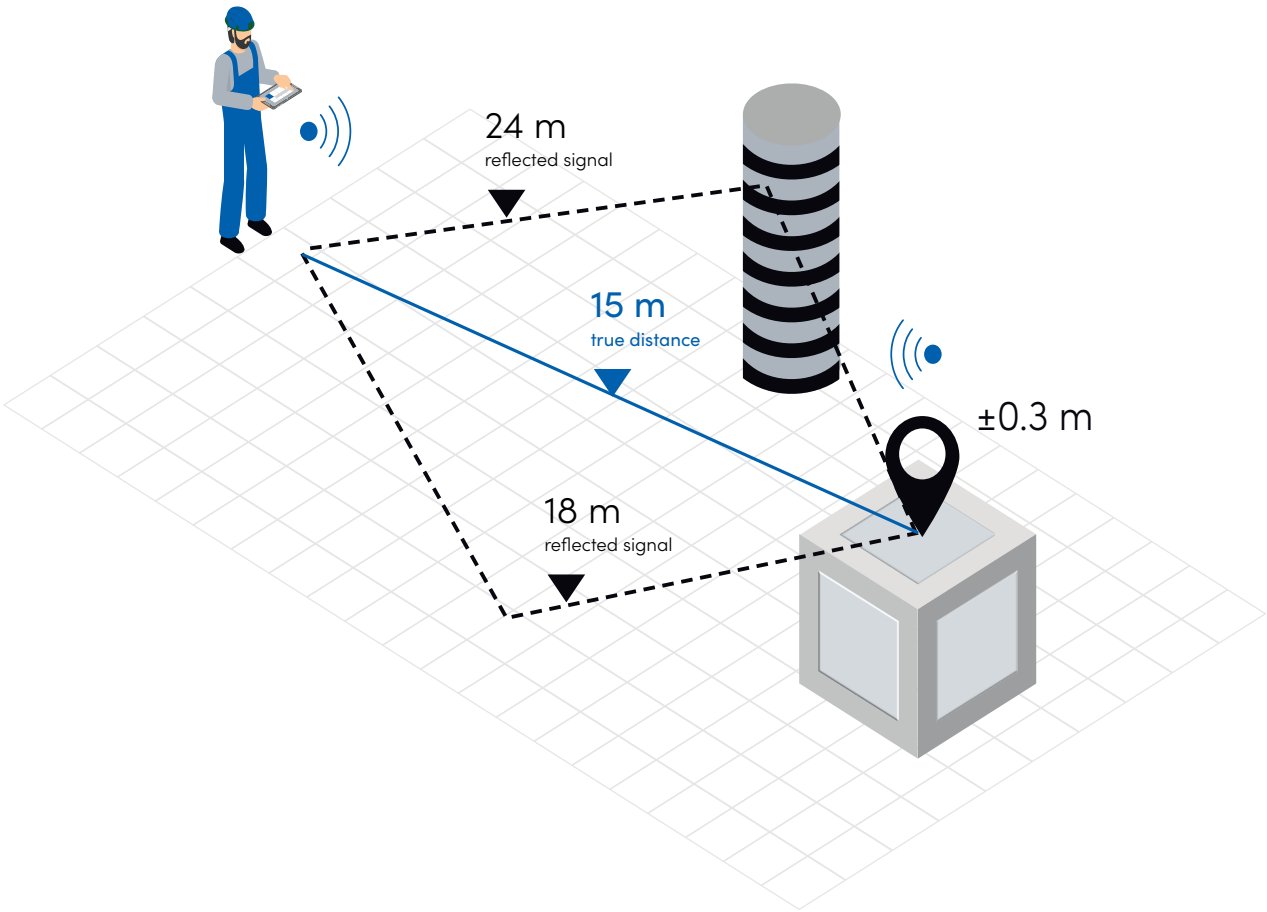
This allows UWB to measure distances with unmatched accuracy, ideal for demanding industrial use cases such as assembly-line automation, high-precision tracking, and safety monitoring.

Multipath Resistance

A common challenge in indoor environments is multipath signal propagation. This occurs when radio signals bounce off walls, machinery, and other surfaces. It's especially common in industrial spaces filled with metal and moving equipment.

Imagine speaking to a friend across a football pitch. Your voice travels directly. If you're on a hill with trees and buildings, your voice will bounce off them and reach the friend distorted or delayed. Radio signals behave the same way.

UWB solves this with short signal pulses (under 2 nanoseconds). These fast pulses allow the receiver to detect the direct path and ignore reflections, reducing positioning errors. Ideal for real-world industrial environments.

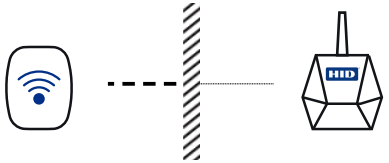


How UWB Handles Line-of-Sight Conditions

LOS
Line-of-sight

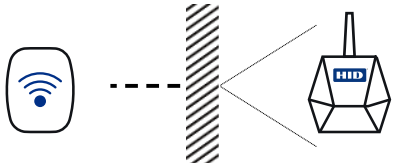
Receivers pick up strong signals that travel straight to them. Reflected signals are ignored.

PLOS
Partial Line-of-sight



Objects can block the direct signal, but the receiver can still catch it, even if it's weaker or slightly distorted, and ignore any reflected signals.

NLOS
Non Line-of-sight



If the direct signal can't reach the receiver, UWB's wide bandwidth helps filter out all reflected signals, so it doesn't give a wrong result.

Location Methods and Topology

UWB relies on precise Time of Flight (ToF) measurements to calculate distance. By measuring how long it takes a pulse to travel from the transmitter to the receiver, UWB can determine accurate positions.

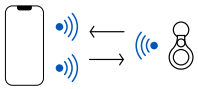
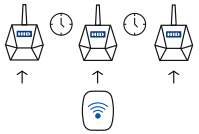
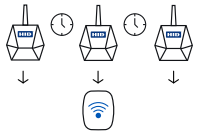
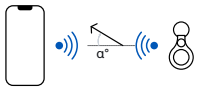
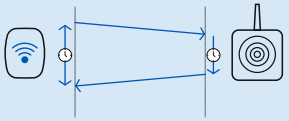
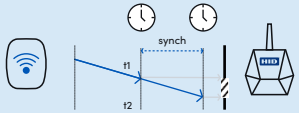
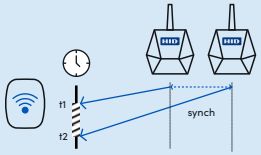
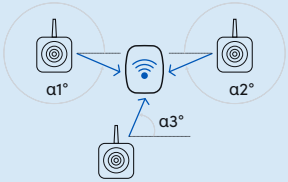
Even amongst other ToF-based systems, UWB has a major advantage. Thanks to its wide bandwidth, it performs reliably even in dense and metallic environments.

Two primary methods used in UWB location systems:

- **Two-Way Ranging (TWR):** Measures the time it takes for a signal to travel back and forth between two devices to calculate their distance.
- **Time Difference of Arrival (TDoA):** Calculates position by comparing the arrival times of signals received at multiple anchor points.

A third method, **Angle of Arrival (AoA),** compares the direction of signals coming to several receivers. It is typically used as a complementary technique to ToF. However, AoA requires a clean line-of-sight, and its accuracy drops with distance. For this reason, it's mainly applied in short-range, less demanding consumer use cases (under 10 meters).

Comparison of location methods

Location methods	Two Way Ranging (TWR)	Uplink Time Difference of Arrival (UL-TDoA)	Downlink Time Difference of Arrival (DL-TDoA)	Angle of Arrival (AoA)
How devices communicate	 Range is calculated at mobile device	 Location is calculated at server	 Location is calculated at tag	 Calculated at mobile device
Signal exchange process				
Key applications	Consumer: mobile driven interactions: home, access, public transport	Industrial RTLS: material handling units, and workers.	Consumer: navigation	Consumer: mobile driven interactions
How position is calculated	Time of Flight	Time of Flight	Time of Flight	Angle of Arrival
Implementation structure	peer-to-peer ranging	infrastructure-based	infrastructure-based	peer-to-peer ranging
Environment criteria	partial line of sight	partial line-of-sight	partial line-of-sight	strict line of sight needed
Tag battery life	days to months	years	days	years
Tag density	tens to hundreds	thousands	unlimited	thousands
RTLS scalability	limited	1000+ anchors	1000+ anchors	1000+ anchors
Interoperability, standards, support	Omlox, FIRA, IEEE 802.15.4z	Omlox, FIRA, IEEE 802.15.4z	Omlox, FIRA, IEEE 802.15.4z	FIRA, IEEE 802.15.4z

How Is UWB Different from Other Indoor Positioning Technologies?

UWB, Bluetooth (BLE) and Wi-Fi are the three dominant wireless technologies used for indoor location tracking. Other alternatives, like ultrasound or infrared, lack key factors for large-scale adoption, including open standards, market maturity, and support from more vendors.

UWB

- Delivers high-precision positioning with centimeter-level accuracy
- Measures time-of-flight of ultra-short, low-power pulses between devices
- Highly resistant to interference and multipath effects
- Performs reliably in demanding environments like factories and warehouses

BLE

- Provides meter-level resolution
- Uses low-energy beacons or tags to communicate with nearby receivers
- Energy efficient
- Commonly used for asset tracking and people monitoring in office, healthcare, and retail environments

WIFI

- Estimates location by analyzing signals from existing access points
- Delivers lower accuracy (5–15 meters)
- Cost-effective and infrastructure-friendly
- Ideal for presence detection and movement analytics in large venues or public spaces

Location Technologies by Precision and Industry Use

Location performance of RTLS is heavily dependent on two factors:

1. How good is visibility between tag and infrastructure nodes or anchors
2. How the technology is resistant to multipath propagation

Resistance to multipath propagation is heavily dependent on the core location method:

Core method	Strengths	Technology sweet spot
Time Difference of Arrival (TDoA)	• High precision and reliability • Resistant to multipath • Works in any environment including high-reflective metal industry • Good for partial-line-of-sight and no-line-of-sight scenarios • Exclusively for UWB due to its wide bandwidth, other technologies can't use it.	UWB
Angle of Arrival (AoA)	• Good accuracy for specific conditions: clear line-of-sight, optimal distance (error grows with the distance) • Sensitive to multipath propagation, resulting in low reliability	
Received signal strength indicator (RSSI)	• Simple and low-cost • Works with existing infrastructure • Estimates distance based on signal strength between the tag and anchors • Not very accurate	

Why UWB & TDoA

Time Difference of Arrival (TDoA) is one of the most accurate ways to measure location, especially when used with UWB.

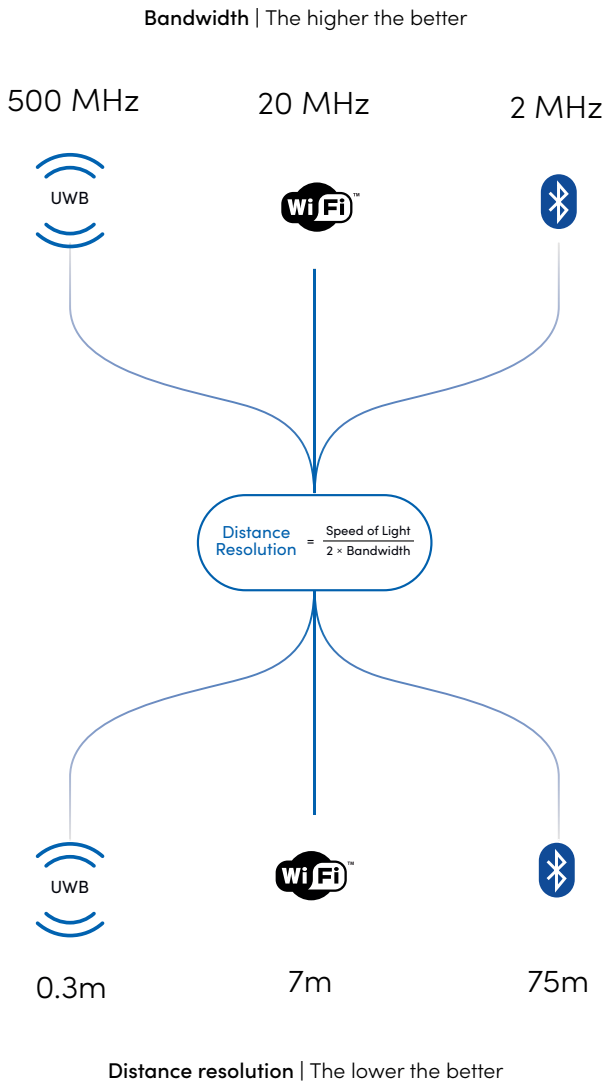
That's because of a simple rule from radar physics: The wider the signal's bandwidth, the more precisely you can measure distance.

Here's the formula:

$$\text{Distance Resolution} = \frac{\text{Speed of Light}}{2 \times \text{Bandwidth}}$$

Narrowband signals like Wi-Fi or Bluetooth don't have enough bandwidth, so their distance measurements are fuzzy. In contrast, UWB has a very wide bandwidth, which lets it measure distances down to just a few centimeters.

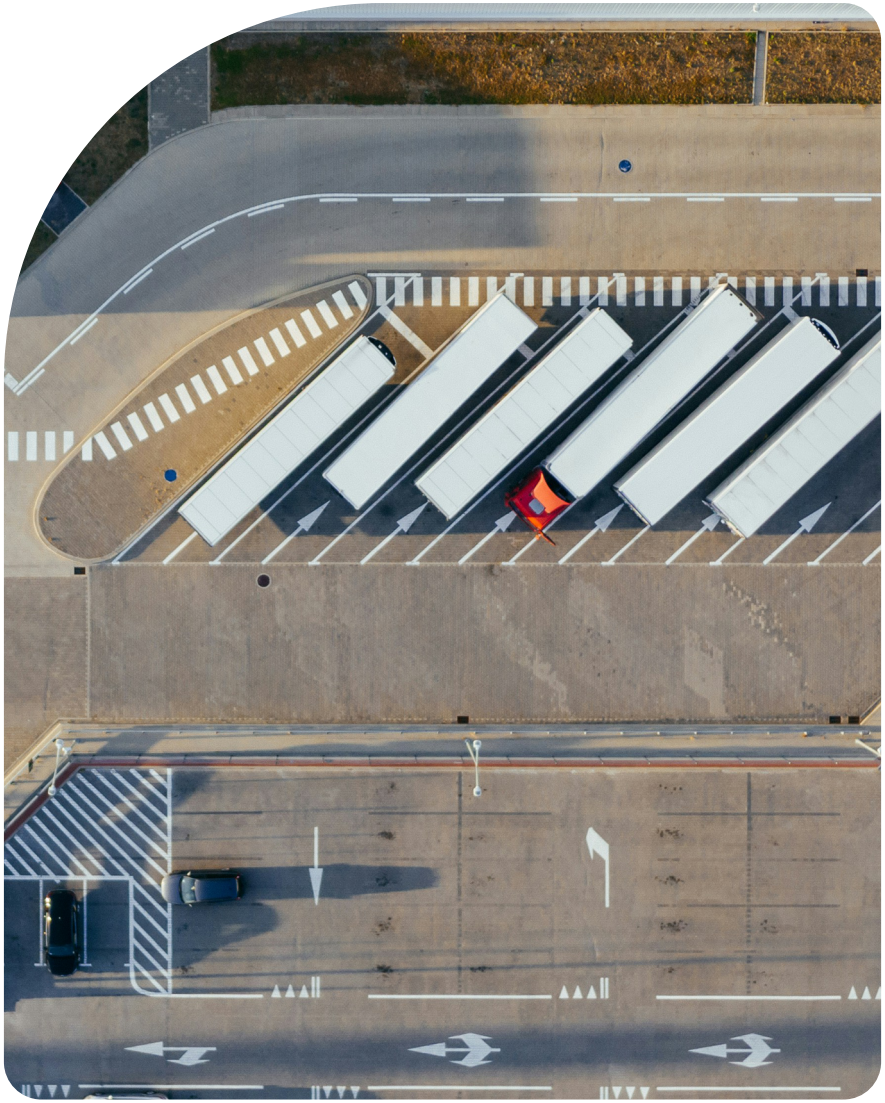
So, when you combine TDoA with UWB, you get high precision and reliable tracking, even in complex industrial environments.



UWB Evolution and Landscape

From custom build to mainstream technology

When large companies assess the introduction of new infrastructure technology, they consider three key layers: chip makers, hardware vendors, and avoiding vendor lock-in by using open standards and ensuring interoperability.



Timeline of UWB Mass Adoption

The story of UWB started around 2007. Back then, there were only two major vendors in the market—and everything was custom-built. It worked, but it didn't scale. Without standards, and with every system locked to a single supplier, real adoption stalled for years.

Then in 2018, Apple announced it was going all-in on UWB. The market exploded. UWB standards brought vendor flexibility, a broader ecosystem, and confidence and Volkswagen introduced the first UWB-powered keyless car entry. Now, there are over **10 million UWB-enabled vehicles.** You can find UWB inside Apple and Samsung smartphones, BMW, Honda, and Volvo cars, Atlas Copco tools, Cisco access points, or Raymond and Linde forklifts.

UWB Standards

There are three key standards important for mass adoption of location technology.
Their work ensures UWB technology is secure, interoperable and ready for global deployment:

omlox

Focus: the world's first open standard for real-time locating systems in industrial environments.

Key technologies: UWB, BLE, GPS

Value offering: Open, vendor-agnostic, multi-tech asset tracking.

Members: 30+ companies, from industrial automation leaders to semiconductor manufacturers.

FiRa Consortium

Focus: FiRa drives the development of UWB technology for secure and accurate location tracking. It ensures reliable performance in real-world conditions, not just in controlled lab environments.

Key technologies: UWB

Value offering: UWB Interoperability, security, and ecosystem

Members: Over 100 organizations, including major players in consumer tech, semiconductors, and networking.

Car Connectivity Consortium (CCC)

Focus: The CCC brings together the automotive and tech sectors to create global standards for secure, seamless device-to-vehicle communication—particularly for digital car keys.

Key technologies: UWB, BT, NFC

Value offering: Secure, interoperable digital keys for vehicles

Members: Over 300 members including top automakers, smartphone manufacturers, chipmakers, and security companies.

Case Studies: What Manufacturers Are Achieving with UWB

Marmon Foodservice Technologies

Enhancing production visibility and efficiency

15% increase in productivity

Marmon needed to increase the productivity of production lines and operators, improve quality control, and base decisions on real-place data. They deployed Sewio UWB RTLS to monitor manufacturing processes with pinpoint accuracy. This helped them identify bottlenecks and streamline workflows and improve overall efficiency.

Prakab

Reducing material consumption and errors in cable production

2% decrease of copper usage

The goal of the digitalization project was to reduce costs by systematically eliminating waste in the company supply chain and production processes. The company implemented RTLS to precisely track all materials and processes. The result is fewer mistakes and better resource management. Plus, the ROI came very quickly—in 12 months.

VELUX Commercial

Transitioning from manual to automated production

10% increase in productivity

VELUX Commercial aimed to enhance production efficiency and safety at their Danish facility. Implementing Sewio's UWB-based RTLS enabled real-time tracking of personnel and equipment, facilitating automated workstation adjustments, indoor navigation, and process optimization through data-driven insights. This transformation resulted in better productivity and a reduction of work-in-progress inventory.

HID Sewio UWB-based RTLS

The Industrial Edge Starts with UWB

UWB doesn't just survive in industrial chaos. It was built for it.

Measuring the time-of-flight—literally the time it takes a signal to travel between two points—gives UWB centimeter-level accuracy for real-time tracking. It works even in some of the toughest places you can imagine: factories full of steel, concrete, and noise. Where GPS fails, UWB thrives.

UWB technology is the foundation of real-time location systems that are already transforming manufacturing. We see our customers successfully using it for automating workflows, quick locating of assets, and safety systems that know where people are in real time. It's precise, reliable, and secure.

And that's what makes using it so exciting! We're no longer guessing, we're measuring. We're not waiting for the future of smart industry. With UWB, it's already here.

About HID's Sewio

Sewio is HID's UWB real-time locating system (RTLS) for precise indoor tracking, improving efficiency, quality, and safety in manufacturing and logistics. After a decade in the market, Sewio joined HID Global, a leader in identity and location services, further strengthening its position in the world of innovative solutions.

- Real-time data and actionable insights
- High accuracy up to 0.3 m (1 ft)
- Up to 99.9% reliability in industrial environments
- One infrastructure for many applications
- More than 300 installations in 45 countries

Ready to dive deeper?
www.hidglobal.com

Lubomír Mráz,
Co-founder of Sewio

About the author

With more than a decade of hands-on experience in IoT and industrial geolocation technologies, Lubomír has shaped several of the innovations that power today's real-time location systems. He holds multiple utility models in this field. He also plays an active role in the omlox community, helping advance interoperability standards that make location services more reliable and accessible for industrial environments.